

Redowanul Akbar

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 Redowanul Akbar |  cse.sds.bracu.ac.bd/faculty_profile/293/redowanul_akbar

OBJECTIVE

Aspiring to a research and teaching career in Computer Science, with a focus on human-centered intelligent systems. My interests include Human-Computer Interaction, Reinforcement Learning, Trustworthy AI, robotics, data science and applied machine learning, with the goal of developing systems that improve human productivity and quality of life while promoting hands-on, experiential learning.

EXPERIENCE

- BRAC University** 

May 2024 - Present
Dhaka, Bangladesh

Lecturer, Department of Computer Science and Engineering

 - Teach undergraduate theory courses of Computer Architecture and Introduction to Robotics as well as labs of Introduction to Robotics and Computer Interfacing.
 - Coordinate the labs of Introduction to Robotics and Computer Interfacing courses as a lab coordinator, manage all lab hardware and software components, lab schedules, operations and instructional delivery.
 - Plan and prepare course study materials and assessments including lectures, lab worksheets, assignments, quizzes, midterms and final examinations.
 - Support students academically through consultation hours, advising and exam invigilation.
 - Actively work as a part of the Outcomes-Based Education (OBE) accreditation team, assisting with the CSE program's accreditation documentation, assessment, and curriculum alignment.
- BRAC University** 

January 2024 - May 2024
Dhaka, Bangladesh

Adjunct Lecturer, Department of Computer Science and Engineering

 - Performed similar duties of teaching, lab coordination, course material and assessment planning and development, student academic support as a Lecturer.
- BRAC University** 

June 2022 - December 2023
Dhaka, Bangladesh

Student Tutor (Teaching Assistant), Department of Computer Science and Engineering

 - Provided consultation hours to support students academically in Programming Language I, Programming Language II and Elements of Statistics and Probability courses.
 - Assisted instructors during lab sessions, to help students with their hands-on coursework.
 - Evaluated weekly lab assignments and coursework under the supervision of faculty members.
 - Supported supervising faculty with academic responsibilities, including examination invigilation and grade sheet preparation.

EDUCATION

- BRAC University**

January 2020 - December 2023
Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering

 - CGPA: 3.90/4.00, Highest Distinction
- Notre Dame College**

May 2017 - June 2019
Dhaka, Bangladesh

Higher Secondary Certificate

 - GPA: 5.00/5.00
- Dhaka Residential Model College**

February 2017
Dhaka, Bangladesh

Secondary School Certificate

 - GPA: 5.00/5.00

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, Assembly Language(x86, RISC-V)
- **Web Technologies:** Express.js, MERN Stack
- **Database Systems:** MySQL, MongoDB
- **Data Science & Machine Learning:** Reinforcement Learning, D3rlpy, Scikit Learn, Numpy, Pandas, OpenAI Gym
- **Other Tools & Technologies:** Microsoft Word, Microsoft Excel, Microsoft PowerPoint, Google Docs, Google Sheets, Google Slides, Canva
- **Specialized Area:** Computer Interfacing, Robotics, Reinforcement Learning, Data Science and Machine Learning, Embedded Systems, Computer Architecture
- **Research Skills:** Problem Solving, Technical Documentation, Data Analysis

PROJECTS

- **Apple Store BD [A collaborative e-commerce website project]** December 2021 - January 2022
Tools: PHP, MySQL, HTML, CSS 
- **TableTop [A collaborative project of table ordering system for restaurants]** March 2022 - May 2022
Tools: PHP, MySQL, HTML, CSS 
- **Mobile Price Classification [A collaborative machine Learning project]** June 2022 - September 2022
Tools: Python, scikit-learn, matplotlib, seaborn 
- **BentoSnaccs: [User-friendly online platform for ordering exotic Japanese snacks]** February 2023 - May 2023
Tools: MERN Stack 
- **Detecting Malicious URLs [A collaborative machine Learning project]** February 2023 - May 2023
Tools: Python, scikit-learn, xgboost, matplotlib, seaborn, Re 

WORKSHOPS

- **Training On STM32 Microcontroller Programming** February 2024 - May 2024
Organized by the Department of Computer Science and Engineering, School of Data and Sciences, BRAC University

RESEARCH WORK

O=ONGOING, T=THESIS

- [T.1] Sumaiya, R. Akbar, U. J. Taposhi, Md. Z. Islam, **A Scalable Bitcoin Blockchain Network by Dynamic Block Size Adjustment using Reinforcement Learning**, B.Sc. thesis, BRAC University, 2023.
- [O.1] **Privacy and Usability Analysis of Humor-Based Nudging**
- [O.2] **Reducing Hallucination of Large Language Models using Reinforcement Learning**

HONORS AND AWARDS

- **Merit-based scholarship** January 2021 - December 2023
BRAC University 
 - A merit-based scholarship for achieving academic excellence during bachelor's degree.
- **VC's List** May 2020 - December 2023
BRAC University 
 - Awarded on the VC's List 7 times during the bachelor's degree for securing outstanding grades.
- **Dean's List** May 2020 - December 2023
BRAC University 
 - Awarded on the Dean's List 3 times during the bachelor's degree for securing outstanding grades.
- **Highest Distinction** December 2023
BRAC University 
 - Awarded with the highest distinction during the completion of the bachelor's degree.

LEADERSHIP EXPERIENCE

- **Lab Coordinator** January 2021 - December 2023
BRAC University
 - Coordinate all lab operations, components and schedules of Computer Interfacing and Introduction to Robotics courses.
 - Develop lab study materials, assessments.
 - Introduced a new microcontroller (*STM32F446RE*) development board to promote register-level programming to the course curriculum of Computer Interfacing

PROFESSIONAL MEMBERSHIPS

- **Human-Centered Computing and Society (HCCS) Research Group** December 2025 - Present
BRAC University 

CERTIFICATIONS

- **DataCamp: Introduction to MLflow** September 2024
- **DataCamp: MLOps Deployment and Life Cycling** September 2024
- **DataCamp: End-to-End Machine Learning** September 2024
- **DataCamp: MLOps Concepts** September 2024
- **DataCamp: Machine Learning for Time Series Data in Python** April 2023

REFERENCES

1. **Dr. Md. Golam Rabiul Alam**

Professor, Department of Computer Science and Engineering
BRAC University
Email: rabiul.alam@bracu.ac.bd
Contact information available on request
Relationship: [Thesis Supervisor]

2. **Dr. Swakkhar Shatabda**

Professor, Department of Computer Science and Engineering
BRAC University
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Contact information available on request
Relationship: [OBE Team Co-ordinator]

3. **Dr. Muhammad Iqbal Hossain**

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BRAC University
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